2.51 Sample Means Hypothesis Testing

Worked Solutions

1 a)

$\bar{X} \sim \mathrm{N}\left(30, \frac{2^2}{4}\right)$

$\mathrm{P}(\bar{X} < 27.1) = 0.0019 < 0.05$

Reject H_0

1 b)

$\bar{X} \sim \mathrm{N}\left(45, \frac{16}{10}\right)$

$\mathrm{P}(\bar{X} < 43.2) = 0.0774 > 0.05$

Fail to reject H_0

1 c)

$\bar{X} \sim \mathrm{N}\left(50, \frac{20}{30}\right)$

$\mathrm{P}(\bar{X} < 49.2) = 0.1640 > 0.1$

Fail to reject H_0

1 d)

$\bar{X} \sim \mathrm{N}\left(85, \frac{37}{15}\right)$

$\mathrm{P}(\bar{X} < 81.2) = 0.0078 < 0.01$

Reject H_0

2 a)

$\bar{X} \sim \mathrm{N}\left(60, \frac{3^2}{5}\right)$

$\mathrm{P}(\bar{X} > 62.1) = 0.0588 > 0.05$

Fail to reject H_0

2 b)

$\bar{X} \sim \mathrm{N}\left(23, \frac{25}{9}\right)$

$\mathrm{P}(\bar{X} > 25.9) = 0.0409 < 0.05$

Reject H_0

2 c)

$\bar{X} \sim \mathrm{N}\left(104, \frac{35}{20}\right)$

$\mathrm{P}(\bar{X} > 107) = 0.0117 < 0.1$

Reject H_0

2 d)

$\bar{X} \sim \mathrm{N}\left(162, \frac{54}{25}\right)$

$\mathrm{P}(\bar{X} > 166.3) = 0.0017 < 0.01$

Reject H_0

3 a)

$\bar{X} \sim \mathrm{N}\left(34, \frac{6^2}{16}\right)$

$\mathrm{P}(\bar{X} < 30) = 0.0038$

p-value $= 2 \times 0.0038 \ldots = 0.0077 < 0.05$

Reject H_0

3 b)

$\bar{X} \sim \mathrm{N}\left(70, \frac{2.25}{7}\right)$

$\mathrm{P}(\bar{X} > 70.8) = 0.0791$

p-value $= 2 \times 0.0791 \ldots = 0.1582 > 0.05$

Fail to reject H_0

3 c)

$\bar{X} \sim \mathrm{N}\left(125, \frac{80}{100}\right)$

$\mathrm{P}(\bar{X} > 126.6) = 0.0368$

p-value $= 2 \times 0.0368 \ldots = 0.0736 < 0.1$

Reject H_0

3 d)

$\bar{X} \sim \mathrm{N}\left(216, \frac{17}{30}\right)$

$\mathrm{P}(\bar{X} < 214) = 0.0039$

p-value $= 2 \times 0.0039 \ldots = 0.0079 < 0.01$

Reject H_0

4 a)

$\bar{X} \sim \mathrm{N}\left(28, \frac{2.5^2}{18}\right)$

CR is $\{\bar{X} < 27.03076\}$

$26.7 < 27.03076$

Reject H_0

4 b)

$\bar{X} \sim \mathrm{N}\left(32, \frac{25}{26}\right)$

CR is $\{\bar{X} < 30.38709\}$

$30.5 > 30.38709$

Fail to reject H_0

4 c)

$\bar{X} \sim \mathrm{N}\left(106, \frac{60}{33}\right)$

CR is $\{\bar{X} < 104.272\}$

$104.4 > 104.272$

Fail to reject H_0

4 d)

$\bar{X} \sim \mathrm{N}\left(340, \frac{129}{80}\right)$

CR is $\{\bar{X} < 337.0459\}$

$336.6 < 337.0459$

Reject H_0

5 a)

$\bar{X} \sim \mathrm{N}\left(75, \frac{10^2}{19}\right)$

CR is $\{\bar{X} > 78.77355\}$

$79.3 > 78.77355$

Reject H_0

5 b)

$\bar{X} \sim \mathrm{N}\left(160, \frac{121}{33}\right)$

CR is $\{\bar{X} > 163.1497\}$

$162.7 < 163.1497$

Fail to reject H_0

5 c)

$\bar{X} \sim \mathrm{N}\left(1250, \frac{3.2}{4}\right)$

CR is $\{\bar{X} > 1251.146\}$

$1252.3 > 1251.146$

Reject H_0

5 d)

$\bar{X} \sim \mathrm{N}\left(670, \frac{163}{60}\right)$

CR is $\{\bar{X} > 673.8344\}$

$674.1 > 673.8344$

Reject H_0

6 a)

$\bar{X} \sim \mathrm{N}\left(85, \frac{7^2}{54}\right)$

CR is $\{\bar{X} < 83.13298\} \cup \{\bar{X} > 86.86702\}$

$82.9 < 83.13298$

Reject H_0

6 b)

$\bar{X} \sim N\left(1.6, \frac{.04}{43}\right)$

CR is $\{\bar{X} < 1.540222\} \cup \{\bar{X} > 1.659778\}$

$1.68 > 1.659778$

Reject H_0

6 c)

$\bar{X} \sim N\left(57, \frac{29}{10}\right)$

CR is $\{\bar{X} < 54.19892\} \cup \{\bar{X} > 59.80108\}$

$54.04 < 54.19892$

Reject H_0

6 d)

$\bar{X} \sim N\left(890, \frac{250}{9}\right)$

CR is $\{\bar{X} < 876.4242\} \cup \{\bar{X} > 903.5758\}$

$876.4242 < 902.9 < 903.5758$

Fail to reject H_0

7 a)

$\bar{X} \sim N\left(35, \frac{8^2}{6}\right)$

$Z = \frac{29.7-35}{\frac{8}{\sqrt{6}}} = -1.622786955$

$P(Z < a) = 0.05 \Rightarrow a = -1.644854$

$-1.622786955 > -1.644854$

Fail to reject H_0

7 b)

$\bar{X} \sim N\left(60, \frac{50}{10}\right)$

$Z = \frac{55.9-60}{\sqrt{\frac{50}{10}}} = -1.833575742$

$P(Z < a) = 0.05 \Rightarrow a = -1.644854$

$-1.833575742 < -1.644854$

Reject H_0

7 c)

$\bar{X} \sim N\left(12.5, \frac{1.4}{4}\right)$

$Z = \frac{11.9-12.5}{\sqrt{\frac{1.4}{4}}} = -1.014185106$

$P(Z < a) = 0.1 \Rightarrow a = -1.281552$

$-1.014185106 > -1.281552$

Fail to reject H_0

7 d)

$\bar{X} \sim N\left(496, \frac{244}{45}\right)$

$Z = \frac{490.1-496}{\sqrt{\frac{244}{45}}} = -2.533747629$

$P(Z < a) = 0.01 \Rightarrow a = -2.326348$

$-2.533747629 < -2.326348$

Reject H_0

8 a)

$\bar{X} \sim N\left(300, \frac{20^2}{3}\right)$

$Z = \frac{319.3-300}{\sqrt{\frac{20^2}{3}}} = 1.671429029$

$P(Z > a) = 0.05 \Rightarrow a = 1.644854$

$1.671429029 > 1.644854$

Reject H_0

8 b)

$\bar{X} \sim N\left(133, \frac{1.21}{5}\right)$

$Z = \frac{133.7-133}{\sqrt{\frac{1.21}{5}}} = 1.422952349$

$P(Z > a) = 0.05 \Rightarrow a = 1.644854$

$1.422952349 < 1.644854$

Fail to reject H_0

8 c)

$\bar{X} \sim N\left(8.8, \frac{.56}{12}\right)$

$Z = \frac{9.085-8.8}{\sqrt{\frac{.56}{12}}} = 1.319293642$

$P(Z > a) = 0.1 \Rightarrow a = 1.281552$

$1.319293642 > 1.281552$

Reject H_0

8 d)

$\bar{X} \sim N\left(8550, \frac{560}{35}\right)$

$Z = \frac{8560.3-8550}{\sqrt{\frac{560}{35}}} = 2.575$

$P(Z > a) = 0.01 \Rightarrow a = 2.326348$

$2.575 > 2.326348$

Reject H_0

9 a)

$\bar{X} \sim \mathrm{N}\left(11, \frac{1}{8}\right)$

$Z = \frac{10.25-11}{\sqrt{\frac{1}{8}}} = -2.121320344$

$\mathrm{P}(Z < a) = 0.025 \Rightarrow a = -1.959964$

$\mathrm{P}(Z > a) = 0.025 \Rightarrow a = 1.959964$

$-2.121320344 < -1.959964$

Reject H_0

9 b)

$\bar{X} \sim \mathrm{N}\left(77, \frac{35}{22}\right)$

$Z = \frac{78.9-77}{\sqrt{\frac{35}{22}}} = 1.506367438$

$\mathrm{P}(Z < a) = 0.025 \Rightarrow a = -1.959964$

$\mathrm{P}(Z > a) = 0.025 \Rightarrow a = 1.959964$

$-1.959964 < 1.506367438 < 1.959964$

Fail to reject H_0

9 c)

$\bar{X} \sim \mathrm{N}\left(170, \frac{560}{6}\right)$

$Z = \frac{154.8-170}{\sqrt{\frac{560}{6}}} = -1.573349475$

$\phi^{-1}(0.05) = -1.644854$

$\phi^{-1}(0.95) = 1.644854$

$\mathrm{P}(Z < a) = 0.05 \Rightarrow a = -1.644854$

$\mathrm{P}(Z > a) = 0.05 \Rightarrow a = 1.644854$

$-1.644854 < -1.573349475 < 1.644854$

Fail to reject H_0

9 d)

$\bar{X} \sim \mathrm{N}\left(35600, \frac{19500}{85}\right)$

$Z = \frac{35552-35600}{\sqrt{\frac{19500}{85}}} = -3.169081401$

$\mathrm{P}(Z < a) = 0.005 \Rightarrow a = -2.575829$

$\mathrm{P}(Z > a) = 0.005 \Rightarrow a = 2.575829$

$-3.169081401 < -2.575829$

Reject H_0

10 a)

Molly believes that the machine is dispensing less soda than it should be, but the alternative hypothesis is $\mu >$ 380, when it should be $\mu <$ 380. Molly finds the correct p-value and compares it with the significance level, but she has come to the wrong

conclusion. As $0.0466 < 0.1$, she should reject H_0, rather than fail to reject it.

10 b)

Let X be the volume of soda dispensed by the machine in millilitres.

$H_0 : \mu = 380$

$H_1 : \mu < 380$

$\bar{X} \sim N\left(380, \frac{3^2}{15}\right)$

$P(\bar{X} < 378.7) = 0.0466$

$0.0466 < 0.1$ so we reject H_0

There is sufficient evidence to suggest that the population mean amount of soda is less than expected.

11)

Let X be the height of a plant over the age of 2 years.

$H_0 : \mu = 130$

$H_1 : \mu > 130$

$\bar{X} \sim N\left(130, \frac{20^2}{10}\right)$

$P(\bar{X} > 138) = 0.1030$

$0.1030 > 0.03$ so we fail to reject H_0

There is insufficient evidence to suggest that the population mean height of plants over the age of 2 years is greater than 130 cm.

12)

Let X be the weekly household spend in pounds.

$H_0 : \mu = 589.60$

$H_1 : \mu \neq 589.60$

$\bar{X} \sim N\left(589.6, \frac{120.30^2}{150}\right)$

$P(\bar{X} > 605) = 0.0585$

$0.0585 > 0.05$ so we fail to reject H_0

There is insufficient evidence to suggest that the population mean weekly household spend has changed.

13)

Let X be the weights of the chocolate bars.

$H_0 : \mu = 100$

$H_1 : \mu \neq 100$

$\bar{X} \sim N\left(100, \frac{5^2}{16}\right)$

$P(\bar{X} < 97.925) = 0.0485$

$0.0485 > 0.025$ so we fail to reject H_0

There is insufficient evidence to suggest that the population mean weight of the

chocolate bars is different to 100 grams.

14 a)

The bakery wants to test whether shortening the time in the oven will reduce the weight of the loaves, but their alternative hypothesis is $\mu \neq 400$ when it should be $\mu < 400$. The second error is that the sample variance is $\frac{6^2}{\sqrt{50}}$ when it should be $\frac{6^2}{50}$.

14 b)

Let X be the weight of baked loaves in grams

$\mathrm{H}_0 : \mu = 400$

$\mathrm{H}_1 : \mu < 400$

$\bar{X} \sim \mathrm{N}\left(400, \frac{6^2}{50}\right)$

$\mathrm{P}(X < a) = 0.05 \Rightarrow a = 398.6043$

$398 < 398.6043$ so we reject H_0

There is sufficient evidence to suggest that the population mean weight of loaves of bread has been reduced.

15 a)

$\bar{X} \sim \mathrm{N}\left(161, \frac{9^2}{60}\right)$

CR is $\{\bar{X} < 158.723\} \cup \{\bar{X} > 163.277\}$

15 b)

Let X be the height of female adults in the UK.

$\mathrm{H}_0 : \mu = 161$

$\mathrm{H}_1 : \mu \neq 161$

$\bar{X} \sim \mathrm{N}\left(161, \frac{9^2}{60}\right)$

CR is $\{\bar{X} < 158.723\} \cup \{\bar{X} > 163.277\}$

$158.723 < 162.4 < 163.277$ so we fail to reject H_0

There is insufficient evidence to suggest that the population mean height of female adults in the UK is different from that in 1998.

16 a)

$\bar{X} \sim \mathrm{N}\left(45, \frac{6^2}{12}\right)$

CR is $\{\bar{X} < 42.15103\}$

16 b)

Let X be the waiting time at the veterinary surgery for those without an appointment in minutes.

$\mathrm{H}_0 : \mu = 45$

$\mathrm{H}_1 : \mu < 45$

$\bar{X} \sim \mathrm{N}\left(45, \frac{6^2}{12}\right)$

CR is $\{\bar{X} < 42.15103\}$

$\bar{x} = 42.41667$

$42.41667 > 42.15103$ so we fail to reject H_0

There is insufficient evidence to suggest that the population mean waiting time at the veterinary surgery for those without an appointment has decreased.

17 a)

The first mistake is that the sample means should be normally distributed with a mean of 7 hours and not 7.3 hours. The second mistake is that because we are looking at the top tail, we want to find the value of a such that there is 0.5% at the top, and not 99.5% at the top.

17 b)

Let X be the number of hours of sleep an adult gets.

$H_0 : \mu = 7$

$H_1 : \mu \neq 7$

$\bar{X} \sim N\left(7, \frac{1.1^2}{80}\right)$

$Z = \frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}} = \frac{7.3-7}{\frac{1.1}{\sqrt{80}}} = 2.439$

$P(Z > a) = 0.005 \Rightarrow a = 2.576$

$2.439 < 2.576$ so we fail to reject H_0

There is insufficient evidence to suggest that the population mean number of hours that adults sleep has changed.

18 a)

$Z = \frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}} = \frac{35-30}{\frac{6}{\sqrt{10}}} = 2.635231383$

$P(Z > a) = 0.01 \Rightarrow a = 2.326348$

18 b)

Let X be the response time for the medication to relieve symptoms.

$H_0 : \mu = 30$

$H_1 : \mu > 30$

$\bar{X} \sim N\left(30, \frac{6^2}{10}\right)$

$Z = \frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}} = \frac{35-30}{\frac{6}{\sqrt{10}}} = 2.635231383$

$P(Z > a) = 0.01 \Rightarrow a = 2.326348$

$2.635231383 > 2.326348$ so we reject H_0

There is sufficient evidence to suggest that the population mean time for the medication to relieve symptoms is longer than 30 minutes.

19 a)

$Z = \frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}} = \frac{38.75-40}{\frac{2}{\sqrt{15}}} = -2.420614591$

$\mathrm{P}(Z < a) = 0.02 \Rightarrow a = -2.053749$

19 b)

Let X be the fuel efficiency of a new car.

$\mathrm{H}_0 : \mu = 40$

$\mathrm{H}_1 : \mu < 40$

$\bar{X} \sim \mathrm{N}\left(40, \frac{2^2}{15}\right)$

$Z = \frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}} = \frac{38.75-40}{\frac{2}{\sqrt{15}}} = -2.420614591$

$\mathrm{P}(Z < a) = 0.02 \Rightarrow a = -2.053749$

$-2.420614591 < -2.053749$ so we reject H_0

There is sufficient evidence to suggest that the population mean fuel efficiency of the car is less than 40 miles to the gallon.

20)

Let X be the weight-loss of an adult member of the fitness club.

$\mathrm{H}_0 : \mu = 8.5$

$\mathrm{H}_1 : \mu \neq 8.5$

$\bar{X} \sim \mathrm{N}\left(8.5, \frac{1.8^2}{15}\right)$

$\mathrm{P}(\bar{X} > 9.2) = 0.0660$

$0.0660 > 0.015$ so we fail to reject H_0

There is insufficient evidence to suggest that the population mean weight-loss of adult members of the fitness club is different to 8.5 kilograms.